Modicon M580 Safety
Standards and Certifications

09/2017

EIO0000002750.00

www.schneider-electric.com

The information provided in this documentation contains general descriptions and/or technical characteristics of the performance of the products contained herein. This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications. It is the duty of any such user or integrator to perform the appropriate and complete risk analysis, evaluation and testing of the products with respect to the relevant specific application or use thereof. Neither Schneider Electric nor any of its affiliates or subsidiaries shall be responsible or liable for misuse of the information contained herein. If you have any suggestions for improvements or amendments or have found errors in this publication, please notify us.

You agree not to reproduce, other than for your own personal, noncommercial use, all or part of this document on any medium whatsoever without permission of Schneider Electric, given in writing. You also agree not to establish any hypertext links to this document or its content. Schneider Electric does not grant any right or license for the personal and noncommercial use of the document or its content, except for a non-exclusive license to consult it on an "as is" basis, at your own risk. All other rights are reserved.

All pertinent state, regional, and local safety regulations must be observed when installing and using this product. For reasons of safety and to help ensure compliance with documented system data, only the manufacturer should perform repairs to components.

When devices are used for applications with technical safety requirements, the relevant instructions must be followed.

Failure to use Schneider Electric software or approved software with our hardware products may result in injury, harm, or improper operating results.

Failure to observe this information can result in injury or equipment damage.

© 2017 Schneider Electric. All Rights Reserved.

Table of Contents

Safety Information 5
About the Book 7
Introduction 9
Protective Treatment of the Modicon M580 Safety Platform 10
Characteristics 10
Environment Tests 11

Safety Information

Important Information

NOTICE

Read these instructions carefully, and look at the equipment to become familiar with the device before trying to install, operate, service, or maintain it. The following special messages may appear throughout this documentation or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a “Danger” or “Warning” safety label indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

⚠ DANGER

DANGER indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

⚠ WARNING

WARNING indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

⚠ CAUTION

CAUTION indicates a hazardous situation which, if not avoided, **could result** in minor or moderate injury.

NOTICE

NOTICE is used to address practices not related to physical injury.

PLEASE NOTE

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction and operation of electrical equipment and its installation, and has received safety training to recognize and avoid the hazards involved.

About the Book

At a Glance

Document Scope

This document presents the Standards and Certifications for the M580 Safety range.

For any information related to functional safety refer to the *M580 Safety Manual*, and *M580 Safety System Planning Guide*.

Validity Note

This document is valid for Unity Pro 13.0 XLS or later.

Related Documents

Title of Documentation	Reference Number
M580 Safety Manual	QGH46982 (English), QGH46983 (French), QGH46984 (German), QGH46985 (Italian), QGH46986 (Spanish), QGH46987 (Chinese)
M580 Safety System Planning Guide	QGH60283 (English), QGH60284 (French), QGH60285 (German), QGH60286 (Spanish), QGH60287 (Italian), QGH60288 (Chinese)

You can download these technical publications and other technical information from our website at http://www.schneider-electric.com/en/download

Product Related Information

⚠ WARNING

UNINTENDED EQUIPMENT OPERATION

The application of this product requires expertise in the design and programming of control systems. Only persons with such expertise are allowed to program, install, alter, and apply this product.

Follow all local and national safety codes and standards.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Introduction

The Modicon M580 Safety platform has been developed to comply with the principal national and international standards concerning electronic equipment for industrial automation systems.

- Requirements specific to programmable controllers: functional characteristics, immunity, resistance, safety, etc.: IEC/EN 61131-2 and IEC/EN/UL/CSA 61010-2-201
- Requirements specific to power utility automation systems: IEC/EN 61000-6-5, IEC/EN 61850-3
- Merchant navy requirements of the major international organizations: unified in IACS (International Association of Classification Societies)
- Compliance with European Directives for CE marking:
  - Electromagnetic Compatibility: 2014/30/EU
  - Machinery: 2006/42/EC
- Ex areas:
  - For USA and Canada: Hazardous location class I, division 2, groups A,B,C, and D
  - For other countries: CE ATEX (2014/34/EU) or IECEx in defined atmosphere Zone 2 (gas) and/or Zone 22 (dust)
    Up-to-date information on which certifications have been obtained is available on our website.

Modicon M580 Safety platform is considered as open equipment and is designed for use in industrial environments, in pollution degree 2, over-voltage category II (IEC 60664-1), and in low-voltage installations, where the main power branch is protected on both wires by devices such as fuses or circuit breakers limiting the current to 15 A for North America and 16 A for the rest of the world.

Protective Treatment of the Modicon M580 Safety Platform

The Modicon M580 Safety platform meets the requirements of TC treatment (Treatment for all Climates).

For installations in industrial production workshops or environments corresponding to TH treatment (Treatment for hot and humid environments), Modicon M580 Safety modules must be embedded in enclosures with minimum IP 54 protection.

The Modicon M580 Safety platforms offer protection to IP 20 level and protection against pins (enclosed equipment). In cases where a position is not occupied by a module, a BMXXEM010 protective cover must be installed.

They can therefore be installed without an enclosure in reserved-access areas which do not exceed pollution level 2 (control room with no dust-producing machine or activity). Pollution level 2 does not take account of more severe environmental conditions: air pollution by dust, smoke, corrosive or radioactive particles, vapors or salts, attack by fungi, insects, etc.

Characteristics

Service Conditions and Recommendations Relating to The Environment (Temperature/Humidity/Altitude)

Characteristics		Modicon M580 Safety Platform
Temperature	Operation	-25...+60 °C (–13...+140 °F)
	Storage	- 40...+ 85 °C (–40...+185 °F)
Relative humidity (without condensation)	Cyclical humidity	+ 5...+95% up to 55 °C (+131 °F)
	Continuous humidity	+ 5...+93% up to 60 °C (+140 °F)
Altitude	Operation	● 0…2000 m (0...6562 ft): full specification: temperature and isolation ● 2000...5000 m (6562...16404 ft): ❍ temperature approx derating: remove 1 °C every 400 m from maximum operation temperature (–1.8 °F/1312 ft) ❍ isolation: Remove 150 Vdc every 1000 m (150 Vdc/3281 ft) For accurate temperature derating calculation, refer to IEC61131-2 Ed4.0 Annex A.

Harsh Environments

Ability to withstand corrosive gases environments, see details hereafter *(see page 18)*.

Supply Voltage

Characteristics		BMECPS4002S Modicon M580 Power Supply
Power Supply	Nominal voltage	100...240 Vac
	Limit voltages	85...264 Vac
	Nominal frequencies	50 or 60 Hz
	Limit frequencies	47 or 63 Hz

Environment Tests

Overview

Standards and levels are given for the following tests:

- Immunity to low frequency interference *(see page 11)* (CE)
- Immunity to high frequency interference *(see page 13)* (CE)
- Electromagnetic emissions *(see page 14)* (CE)
- Immunity to climatic variations (Power ON) *(see page 15)*
- Withstand to climatic variations (Power OFF) *(see page 16)*
- Immunity to mechanical constraints (Power ON) *(see page 16)*
- Withstand to mechanical constraints (Power ON) *(see page 17)*
- Equipment and personnel safety *(see page 17)* (CE)
- Specific environment *(see page 18)*

NOTE: The tests with (CE) marking are required by European directives and based on IEC/EN 61131-2 standards.

Immunity to Low Frequency Interference (CE) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Voltage and frequency variations	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-11	0.85...1.10 Un –0.94...1.04 Fn; 4 steps t = 30 min
	IACS E10; IEC 61000-4-11	0.80 Un...0.90 Fn; 1.20 Un...1.10 Fn; t = 1.5 s/5 s
Where: **PS1** applies to PLC supplied by battery **PS2** applies to PLC energized from AC or DC supplies **Un** nominal voltage **Fn** nominal frequency **Udl** detection level when powered		

Name of test	Standards	Levels
Direct voltage variations	IEC/EN 61131-2; IEC 61000-4-29; IACS E10 (PLC not connected to charging battery)	0.85...1.2 Un + ripple: 5% peak; 2 steps t = 30 min
Third harmonic	IEC/EN 61131-2	H3 (10% Un), 0°/180°; 2 steps t = 5 min
Immunity to conducted low frequency (only IACS)	IACS E10	For AC: ● H2…H15 (10% Un), H15…H100 (10%...1% Un), H100…H200 (1% Un) For DC: ● H100…H200 (1% Un)
Voltage interruptions	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-11; IEC 61000-4-29; IACS E10 For functional safety (DS criteria): IEC 61000-6-7; IEC 61326-3-1	Power supply immunity: ● 10 ms for AC and DC PS2 (20 ms DS criteria) ● Check operating mode for longer interruptions up to 5 s, 85% Un For IACS, 3 times 30 s in 5 mn, 85% Un
	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-11	For AC PS2: ● 20% Un, t0: ½ period ● 40% Un, cycle: 10/12 ● 70% Un, cycle: 25/30 ● 0% Un, cycle: 250/300
Voltage shut-down and start-up	IEC/EN 61131-2	● Un…0…Un; t = Un/60 s ● Umin…0…Umin; t = Umin/5 s ● Umin…0.9 Udl…Umin; t = Umin/60 s
Magnetic field	IEC/EN 61131-2; IEC 61000-4-8 (for MV power stations: IEC 61000-6-5; IEC 61850-3) For functional safety (DS criteria): IEC 61000-6-7; IEC 61326-3-1	Power frequency: 50/60 Hz, 100 A/m continuous...1000 A/m; t = 3 s; 3 axes
	IEC 61000-4-10	Oscillatory: 100 kHz...1 MHz, 100 A/m; t = 9 s; 3 axes
Conducted common mode disturbances range 0 Hz…150 kHz	IEC 61000-4-16 (for MV power stations: IEC 61000-6-5; IEC 61850-3) For functional safety (DS criteria): IEC 61000-6-7; IEC 61326-3-1	For AC: 10 V For DC: 10 V cont. or 100 V, t = 1 s
Where: **PS1** applies to PLC supplied by battery **PS2** applies to PLC energized from AC or DC supplies **Un** nominal voltage **Fn** nominal frequency **Udl** detection level when powered		

Immunity to High Frequency Interference (CE) Tests

NOTE: These tests are performed without an enclosure, with devices fixed on a metal grid and wired as per the recommendations in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Electrostatic discharges	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-2; IACS E10 For functional safety (DS criteria): IEC 61000-6-7; IEC 61326-3-1	6 kV contact; 8 kV air; 6 kV indirect contact
Radiated radio frequency electromagnetic field	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-3; IACS E10 For functional safety (DS criteria): IEC 61000-6-7; IEC 61326-3-1	80 MHz...1 GHz: 10/15 V/m (20 V/m DS criteria) 1.4 GHz...2 GHz: 3 V/m (10 V/m DS criteria) 2 GHz...6 GHz: 3 V/m Sinus amplitude modulated 80%,1 kHz + internal clock frequencies
Electrical fast transient bursts	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-4; IACS E10 For functional safety (DS criteria): IEC 61000-6-7; IEC 61326-3-1	For AC or DC main supplies: ● 2 kV in common mode/2 kV in wire mode For AC or DC auxiliary supplies, AC unshielded I/O: ● 2 kV in common mode For analog, DC unshielded I/O, communication and shielded lines: ● 1 kV in common mode
Surge	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-5; IACS E10 For functional safety (DS criteria): IEC 61000-6-7; IEC 61326-3-1	For AC or DC main and auxiliary supplies, AC unshielded I/O: ● 2 kV in common mode/1 kV in differential mode (4 kV DS criteria with external protection) For analog, DC unshielded I/O: ● 2 kV in common mode/2 kV in differential mode For communication and shielded lines: ● 1 kV in common mode (3 kV DS criteria)

Name of test	Standards	Levels
Conducted disturbances induced by radiated electromagnetic fields	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-6; IACS E10 For functional safety (DS criteria): IEC 61000-6-7; IEC 61326-3-1	10 V; 0.15 MHz...80 MHz (20 V DS criteria) Sinus amplitude 80%, 1 kHz + spot frequencies
Damped oscillatory wave	IEC/EN 61131-2; IEC 61000-4-18; IACS E10	For AC or DC main supplies and AC auxiliary supplies, AC unshielded I/O: ● 2.5 kV in common mode/1 kV in differential mode For DC auxiliary supplies, analog, DC unshielded I/O: ● 1 kV in common mode/0.5 kV in differential mode For communication and shielded lines: ● 0.5 kV in common mode

Electromagnetic Emissions (CE) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Conducted emission	IEC/EN 61131-2; IEC/EN 61000-6-4; CISPR 11 and 22, Class A, Group 1 (FCC part 15 compliance)	150…500 kHz: quasi-peak 79 dB (µV/m); average 66 dB (µV/m) 500 kHz…30 MHz: quasi-peak 73 dB (µV/m); average 60 dB (µV/m)
	IACS E10	● AC or DC power (general power distribution zone): 10 kHz…150 kHz: quasi-peak 120...69 dB (µV/m); 150 kHz…0.5 MHz: quasi-peak 79 dB (µV/m) 0.5…30 MHz: quasi-peak 73 dB (µV/m) ● AC or DC power (bridge and deck zone for evaluation): 10 kHz…150 kHz: quasi-peak 96…50 dB (µV/m) 150 kHz…0.35 MHz: quasi-peak 60…50 dB (µV/m) 0.35…30 MHz: quasi-peak 50 dB (µV/m)

Name of test	Standards	Levels
Radiated emission	IEC/EN 61131-2; IEC/EN 61000-6-4; CISPR 11 and 22, Class A, Group 1 (FCC part 15 compliance)	30 MHz...230 MHz: quasi-peak 40 dB (µV/m) (at 10 m); 230 MHz...1 GHz: quasi-peak 47 dB (µV/m) (at 10 m); 1...3 GHz: quasi-peak 76 dB (µV/m) (at 3 m); 3...6 GHz: quasi-peak 80 dB (µV/m) (at 3 m);
	IACS E10	For general power distribution zone: ● 0.15 MHz … 30 MHz: quasi-peak 80…50 dB (µV/m) (at 3 m) ● 30...100 MHz: quasi-peak 60…54 dB (µV/m) (at 3 m) ● 100 MHz...2 GHz: quasi-peak 54 dB (µV/m) (at 3 m) ● 156…165 MHz: quasi-peak 24 dB (µV/m) (at 3 m)

Immunity to Climatic Variations (Power ON) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Dry heat	IEC 60068-2-2 (Bb and Bd)	60 °C (+140 °F), t = 16 hrs
	IACS E10	60 °C (+140 °F), t = 16 hrs +70 °C (+158 °F), t = 2 hrs
Cold	IEC 60068-2-1 (Ab and Ad); IACS E10	-25…0 °C (–13...32 °F), t = 16 hrs + power on at –25 °C
Damp heat, steady state (continuous humidity)	IEC 60068-2-78 (Cab); IACS E10	55 °C (+131 °F), 93% relative humidity, t = 96 hrs (for ruggedized range: 60 °C (+140 °F))
Damp heat, cyclic (cyclical humidity)	IEC 60068-2-30 (Db); IACS E10	25…55 °C (77...131 °F), 93…95% relative humidity, 2 cycles t = 12 hrs + 12 hrs
Change of temperature	IEC 60068-2-14 (Nb)	0 …60 °C (32...140 °F), 5 cycles t = 6 hrs + 6 hrs (for ruggedized range: -25…70 °C (77...158 °F))

Withstand to Climatic Variations (Power OFF) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Dry heat	IEC/EN 61131-2; IEC 60068-2-2 (Bb and Bd); IEC/EN 60945	85 °C (185 °F), t = 96 hrs
Cold	IEC/EN 61131-2; IEC 60068-2-1 (Ab and Ad); IACS E10	–40 °C (–40 °F), t = 96 hrs
Damp heat, cyclic (cyclical humidity)	IEC/EN 61131-2; IEC 60068-2-30 (Db)	25…55 °C (77...131 °F), 93…95% relative humidity, 2 cycles t = 12 hrs + 12 hrs
Change of temperature (thermal shocks)	IEC/EN 61131-2; IEC 60068-2-14 (Na)	–40...85 °C (–40...185 °F), 5 cycles t = 3 hrs + 3 hrs

Immunity to Mechanical Constraints (Power ON) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Sinusoidal vibrations	IEC/EN 61131-2; IEC 60068-2-6 (Fc)	Basic IEC/EN 61131-2: ● 5...150 Hz, ± 3.5 mm amplitude (5…8.4 Hz), 1 g (8.4…150 Hz) Enhanced profile: ● 5...150 Hz, ± 10.4 mm amplitude (5….8.4 Hz), 3 g (8.4….150 Hz) For basic and specific: endurance: ● 10 sweep cycles for each axis
	IEC 60870-2-2; IEC 60068-2-6 (Class Cm) (for telecontrol equipment and system)	3...8.4 Hz, 7 mm amplitude, 8.4...500 Hz, 2 g Endurance: 10 sweep cycles for each axis
	IACS E10	3...100 Hz, 1 mm amplitude (3…13.2 Hz), 0.7 g (13.2 …100 Hz) Endurance at each resonance frequency: 90 min for each axis, amplification coefficient < 10
	IEC 60068-2-6	Sismic analysis: 3...35 Hz, 22.5 mm amplitude (3…8.1 Hz), 6 g (8.1…35 Hz)

Name of test	Standards	Levels
Shocks	IEC/EN 61131-2; IEC 60068-2-27 (Ea)	15 g, 11 ms; 3 shocks/direction/axis
Free fall during operation	IEC/EN 61131-2; IEC 60068-2-32 (Ed Method 1)	1 m (3.28 ft), 2 falls

Withstand to Mechanical Constraints (Power OFF) Tests

Name of test	Standards	Levels
Random free fall with packaging	IEC/EN 61131-2; IEC 60068-2-32 (Method 1)	1 m (3.28 ft), 5 falls
Flat free fall	IEC/EN 61131-2; IEC 60068-2-32 (Ed Method 1)	10 cm (0.33 ft), 2 falls
Controlled free fall	IEC/EN 61131-2; IEC 60068-2-31 (Ec)	30° or 10 cm (0.33 ft), 2 falls
Plugging/Unplugging	IEC/EN 61131-2	For modules and connectors: Operations: 50 for permanent connections, 500 for non-permanent connections

Equipment and Personnel Safety Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Dielectric strength and insulation resistance	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	Dielectric: Min. 2 Un + 1000 Vac (see details in *M580 Safety System Planning Guide*); t = 1 min Insulation: Un ≤ 50 V: 10 MΩ, 50 V ≤ Un ≤ 250 V: 100 MΩ
Continuity of earth	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	30 A, R ≤ 0,1 Ω; t = 2 min
Leakage current	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	< 0.5 mA in normal condition < 3.5 mA in single fault condition
Protection offered by enclosures	IEC/EN 61131-2; IEC61010-2-201;	IP20 and protection against standardized pins
Overload	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	50 cycles, Un, 1.5 In; t = 1 s ON + 9 s OFF
Endurance	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	In, Un; 6000 cycles, 1 s ON + 9 s OFF
Temperature rise	IEC/EN 61131-2; UL; CSA; ATEX; IECEx	Ambient temperature 60 °C (140 °F)

Specific Environment Test

NOTE: Refer also to the chapter *Hardened Equipment*.

Name of test	Standards	Levels
Corrosion areas - gas, salt, dust	ISA S71.4	Flowing mixed gas; class Gx, 25 °C (77 °F), 75% relative humidity, t = 14 days
	IEC/EN 60721-3-3; IEC60068-2-60	Flowing mixed gas; class 3C3, 25 °C (77 °F), 75% relative humidity, t = 14 days
	IEC/EN 60721-3-3; IEC60068-2-60	Flowing mixed gas; class 3C4, 25 °C (77 °F), 75% relative humidity, t = 7 days
	IEC60068-2-52	Salt spray: test Kb, severity 2
	IEC/EN 60721-3-3; IEC60068-2-68	Dust and sand, Arizona dust, class 3S4, 20 cycles
	IEC/EN 60721-3-3; IEC60068-2-10	Mould growth, fungal spore, class 3B2, t = 28 days